ZOO-301 FINAL 2021(by M.Faryad)

- The **veins** which carry blood to capillary system in kidneys constitute the renal portal system.
- The **arteries** are the blood vessels that deliver oxygen-rich blood from the heart to the tissues of the body.
- **Arteries** carry blood loaded with oxygen from your heart to the rest of your body. **Veins** deliver the blood, now without much of the oxygen, back to your heart.
- The foreign substance that elicits the immune response is called an **antigen.**
- The immune system responds to an antigen by producing specific proteins called **antibodies.**
- There are 4 main types of vaccines:

1. Live-attenuated vaccines.
2. Inactivated vaccines.
3. Subunit, recombinant, polysaccharide, and conjugate vaccines.
4. Toxoid vaccines.

coelomic fluid The **fluid** within the **coelom**; in some animal groups it functions as a hydrostatic skeleton and it may also serve as a circulatory medium.

All vertebrates have a closed circulatory system in which the blood never leaves the **blood vessels.**5

I. Blood moves from the heart, through arteries, arterioles, capillaries, venules, veins, and back to the heart.
II. Exchange between the blood and extracellular fluid only occurs at the capillary level.
III. The heart is the center to the circulatory system.
IV. It is fist-sized pump that circulates the blood throughout the body.
V. As animals became more multicellular and complex the circulatory system evolved because simple diffusion was insufficient to supply all of the cells with nutrients.
VI. The coordination of the circulatory system and the respiratory system to ensure proper gas exchange is very important in animals that have lungs and gills.

Respiration in Birds.5

I. Respiration in birds is complex and efficient.
II. It consists of lungs , 5-6 pairs of air sacs and air passages.
III. Trachea divides into right and left bronchi
IV. Bird lungs have tunnel-like passages called parabronchi, which lead to air capillaries in which gas exchange occurs.
V. The end of each bronchus divides into small branches leading into air sacs.
VI. These sacs are collapsible, and open and close as a result of muscle contractions around them.
VII. The arrangement of air sacs make one-way flow possible.

1. Fixed membrane-receptor mechanism.3

I. This complex activates the enzyme-adenylate cyclase in the membrane.
II. The activated enzyme converts ATP into a nucleotide called cyclic AMP.
III. Cyclic AMP diffuses throughout the cytoplasm and activates an enzyme called protein kinase.

2. Mobile-receptor mechanism.3

I. A steroid hormone molecule (e.g., testosterone) diffuses from the blood to a target cell across the plasma membrane of the target cell.
II. Once in the cytoplasm, the hormone binds to a receptor that carries it into the nucleus.
III. This steroid-protein complex triggers transcription of specific gene regions of DNA.

Renal Portal System

I. The veins which carry blood to capillary system in kidneys constitute the renal portal system.
II. The renal portal vein collects blood from all parts of the hind limbs and enters the kidney by number of branches from outer margin.
III. Inside the kidney these veins divide and re divide to form capillaries. These capillaries form 5 to 6 renal veins on emerging from the kidneys and open into post caval vein. All these veins forms renal portal system

The immune system can be divided into three basic lines of defense against pathogenic infection:

- The first line of defense against infection are the surface barriers that prevent the entry of pathogens into the body
- The second line of defense are the <u>non-specific</u> phagocytes and other internal mechanisms that comprise **innate** immunity
- The third line of defense are the <u>specific</u> lymphocytes that produce antibodies as part of the **adaptive** immune response.

NON-SPECIFIC DEFENCES (INNATE IMMUNITY)		SPECIFIC DEFENCES (ADAPTIVE IMMUNITY)
First line of defense	Second line of defense	Third line of defense
• Skin • Mucous membranes • Secretions of skin and mucous membranes	• Phagocytic leukocytes • Antimicrobial proteins • Inflammatory response • Fever	• Lymphocytes • Antibodies • Memory cells

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- **<u>Fibrinogen</u>** is necessary for blood coagulation (clotting), and the
- **<u>Globulins</u>** include the immunoglobulins, which functions in the immune response because it consists mostly of antibodies.
- **<u>Serum</u>** is plasma from which fibrinogens have been removed.
- **Plasma**, the liquid portion of **blood**, which transports nutrients and wastes and maintains **blood** pressure and homeostasis.

The endocrine system is controlled by complex **<u>feedback mechanisms</u>** in which the secretion of a hormone is turned on (stimulated) or turned off (inhibited), depending on its concentration.

<u>Hemocyanin</u> is the most commonly occurring respiratory pigment in molluscs and certain crustaceans.

Heart is enclosed in double membrane sac called **pericardium.**

Chromaffin tissue is a group of neuroendocrine cells in the adrenal medulla that synthesizes steroid hormones.

Exoskeleton in Arthropods Cuticle waterproofs the body. The **cuticle** provides muscular support

Diaphragm, dome-shaped, muscular and membranous structure that separates the thoracic and abdominal cavities in mammals.

Respiratory Surfaces: Protists and animals have five main types of respiratory systems (surfaces):

1. simple diffusion across plasma membranes,
2. tracheae
3. Skin or cutaneous (integument or body surface) exchange,
4. gills
5. lungs.

A **chemical messenger** is any compound that serves to transmit a message, and may refer to: Hormone, long range **chemical messenger**

These chemical messengers can be categorized as: Type

1. Local chemical messengers (lumones),
2. Neurotransmitters (e.g., acetylcholine),
3. Neuropeptides,
4. Hormones
5. Pheromones

Hemoglobin is a reddish pigment that contains iron as the oxygen-binding metal.

Vertebrate lymphatic system.

1. The lymphatic system is part of the vascular system.
2. The system begins with network of lymphatic vessels that carry a clear fluid called lymph
3. The small lymphatic capillaries merge to form larger lymphatic vessels called lymphatics.
4. Lymphatics are thin-walled vessels with valves that ensure the one-way flow of lymph.
5. Lymph is the extracellular fluid that seeps from the bloodstream and accumulates in the lymph vessels.
6. The lymphatic vessels pass through the lymph nodes on their way back to the heart.
7. Lymph nodes concentrate in several areas of the body and act as disposal centers for foreign particles and cellular debris.
8. They play an important role in the body's defense against disease.

What is the endocrine gland?
Endocrine glands are ductless **glands** of the **endocrine** system that secrete their products, hormones, directly into the blood. The major **glands** of the **endocrine** system include the

pineal **gland**, pituitary **gland**, pancreas, ovaries, testes, thyroid **gland**, parathyroid **gland**, hypothalamus and adrenal **glands**.

Thyroid gland:

a) The thyroid gland is located in the neck, anterior to the trachea.
b) Two of its secretions are thyroxine and triiodothyronine, both of which influence overall growth, development, and metabolic rates.

Four major classes of ovarian hormones help to regulate female reproductive functions

1. Estrogens (estrin, estrone, and estradiol).
2. Progestins
3. Relaxin
4. Inhibin

The estrogens **help regulate the menstrual and estrus cycles and the development of the mammary glands** and other female secondary sexual characteristics.

 The progestins (primarily progesterone) also regulate the menstrual and estrus cycles, and the development of mammary glands.

 It also aid in placenta formation during pregnancy.

 Relaxin, which is produced in small quantities, softens the opening of the uterus (cervix) at the time of delivery.

 The ovaries also produce inhibin, which inhibits the secretion of FSH.

Bimodal breathing is the ability of an organism to exchange respiratory gases simultaneously with both air and water. e.g., some salamanders, crabs, barnacles, bivalve molluscs, and lung fishes.

 Bimodal breathing was an important respiratory adaptation with evolutionary transition between aquatic and terrestrial habitats.

The forebrain has two main parts- and **telencephalon** **a**nd **diencephalon**.

Telencephalon consists of :

a. Cerebrum,
b. Olfactory bulb,
c. Limbic system and
d. Corpus Striatum

Lymphocytes are of types of **two types**: B cells and T cells

Any system that supplies the body tissues with oxygen, nutrients, hormones, and removes waste products is called **transport system or circulatory system.**

The **thalamus** relays all the sensory information to higher brain centers.

What is insect cuticle made of?
Insect cuticle is a layered, fibrous composite of chitin, water, protein, catechol, lipid and occasionally metal and mineral, secreted by a single layer of epidermal cells
how arthropod form its new cuticle

Early in the process of apolysis the epithelial cells release enzymatic moulting fluid between the old **cuticle** and the epidermis. The enzymes partly digest the endocuticle and the epidermis absorbs the digested material for the animal to assimilate. Much of that digested material is re-used to build the **new cuticle**.

how is a pearl formed in bivalves

Pearls are formed in molluskan **bivalves** (clams, oysters, mussels) of several species by the secretion of a substance known as nacre around an irritant in the outer tissue (mantle) of the organism, or between the outer tissue and the shell

The air-conducting portion of the respiratory system of air-breathing vertebrates moves air into (inhalation) and out of (exhalation) this system. This process of air movement is called **ventilation**

Blood Portal Systems in Vertebrates

 Portal system is a system of blood vessels that carry blood from one bed of capillaries to the other bed of capillaries

. Three portal systems are common:

a. Hepatic portal system

b. Renal portal system

c. Hypophysial portal System

Hepatic Portal System

 Nutrients from the small intestine and the right half of the colon are absorbed into blood through the bed of capillaries.

 This blood then moves to the liver through the Hepatic Portal Vein to store and metabolize in liver.

 From the liver the blood enters main blood stream via hepatic vein into the inferior vena cava to the heart.

Pumping Action of Human Heart.

1. Adult human heart beats or pumps **72** times a minute.
2. In the beating of the heart the atria contract simultaneously and ventricles immediately after words.
3. The deoxygenated blood from the lower and upper part of the body is brought into the right atrium
4. On the contraction of the ventricles the blood from the right ventricle is taken to the lungs by the pulmonary artery.
5. The valves open and close due to blood pressure changes when the heart contracts during each heartbeat.
6. Like the valves in veins, heart valves keep blood moving in one direction, preventing backflow.

Nature of Hormones

Hormones can be non steroids(composed of amino acids), steroids(made from cholesterol), and eicosanoids

function of kidney hormones Kidneys Produce three hormones: 5

i) Erythropoietin
ii) Urotensin
iii) Calcitrol

i) **Erythropoietin** Stimulates red blood cell production, the target is bone marrow.

ii) **Urotensin** stimulates constriction of the arteries. Vasoconstriction is needed to help maintain healthy blood flow. It keeps our body temperature from getting too cold It can also raise blood pressure when it's necessary.

iii) **Calcitrol** aids in the absorption of dietary calcium and phosphorus from small intestine.

difference between coelom and pseudocoelom

Coelom or true body cavity is lined with peritoneum.

Pseudocoelom or false body cavity is partially lined with peritoneum.

The **coelom** is developed by splitting of the mesoderm.

The **pseudocoelom** is developed from the blastocoel of the embryo.

Passive immunity is conferred from outside the body, so it doesn't require exposure to an infectious agent or its antigen.

BEST OF LUCK